

pedestrians can be integrated in simulation scenarios. It offers traffic management capabilities with video detectors and induction loops, and provides online interaction through its Traffic Control Interface (TraCI).

CupCarbon

URL: <https://www.cupcarbon.com/>

CupCarbon is an open source simulator supporting multiple programming languages so that IoT integrated vehicular networks can be analysed and controlled. The features that enable it to simulate vehicular communication include:

- Wireless simulator for simulation of smart cities
- Smart vehicular network
- Advanced wireless protocols with IoT integration

OMNeT++

URL: <https://omnetpp.org/>

OMNeT++ is a modular, component-based C++ simulation library and framework. It simulates communication networks, internet protocols, performance modelling of network systems, and wireless ad-hoc

networks. It has extensible and modular architecture for building network simulators, and supports wired and wireless communication networks. Integration of on-chip networks and the INET framework helps simulate communication networks.

PTV VISSIM

URL: <https://www.ptvgroup.com/en/products/ptv-vissim>

This microscopic, multimodal traffic flow simulation software helps with:

- Public transport and urban planning
- Evaluation and optimisation of transport infrastructure performance
- Traffic engineering and transportation planning
- Evaluation of traffic signal operations
- Traffic patterns of all road users on a microscopic scale

It analyses traffic congestion, emissions, and road space distribution.

MATSim (Multi-Agent Transport Simulation Toolkit)

URL: <https://matsim.org/>

This open source project for agent-based transport simulations features:

- Large-scale transport simulations
 - Modules for demand modelling
 - Toolbox for large-scale agent-based transport simulations
 - Agent-based mobility studies
 - Modules for re-planning the road network
 - Transportation demand modelling
 - Modules for mobility simulation
- It allows iterative simulation runs and output analysis.

Working with the SUMO simulator

SUMO is free and open source, highly portable, microscopic, and offers a continuous road traffic simulation package designed to handle large road networks. It is widely used for modelling and analysing traffic scenarios in smart city and vehicular network research.

SUMO can be downloaded from the official website. Use the link <https://sumo.dlr.de/docs/Downloads.html>. Its key components are:

Netedit: This graphical user interface (GUI)-based network editor is used to create and modify road networks and

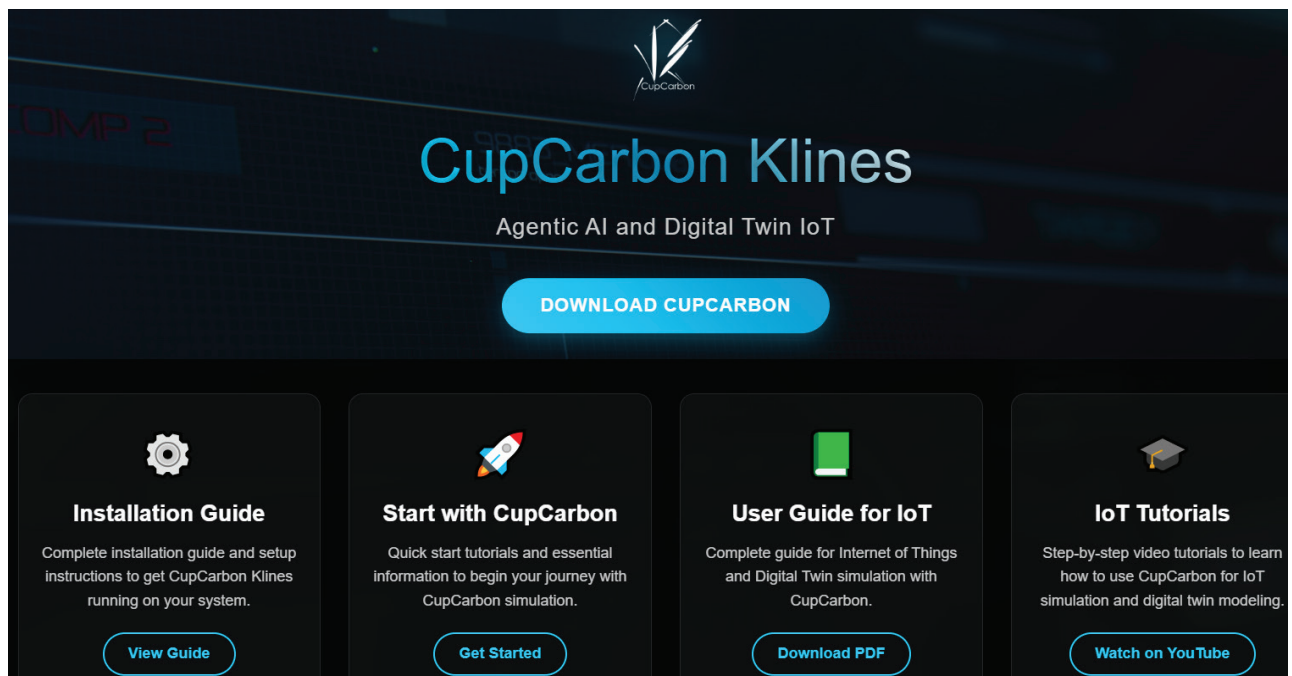


Figure 2: CupCarbon simulator for IoT, wireless and vehicular networks